

- 1) Use the appropriate command to list all files larger than 1 MB in the current directory and save the output to a file.

```
rohit45@rohit45-VirtualBox:~$ ls -lb /var/log | awk ' $5>1048576 {print $0}' >
opfile.txt
```

```
rohit45@rohit45-VirtualBox:~$ cat opfile.txt
```

```
-rw-r--r-- 1 root      root      1176523 Jan 31 12:21 dpkg.log.1
-rw-r----- 1 syslog   adm        1781888 Feb  1 11:53 kern.log.1
-rw-r----- 1 syslog   adm        5840041 Feb  1 11:53 syslog.1
```

- 2) Replace all occurrences of "localhost" with "127.0.0.1" in a configuration file named config.txt, and save the updated file as updated_config.txt.

```
rohit45@rohit45-VirtualBox:~$ vi config.txt
rohit45@rohit45-VirtualBox:~$ cat config.txt
```

```
server_name=localhost,localhost
server_port=8080
```

```
database_host=localhost
database_port=3306
database_user=root
database_password=password
```

```
api_url=http://localhost/api/v1
callback_url=http://localhost/callback
```

```
log_server=localhost
backup_server=localhost
```

```
rohit45@rohit45-VirtualBox:~$ cat config.txt | sed 's/localhost/127.0.0.1/g' >
updated_config.txt
rohit45@rohit45-VirtualBox:~$ cat updated_config.txt
```

```
server_name=127.0.0.1,127.0.0.1
server_port=8080
```

```
database_host=127.0.0.1
database_port=3306
database_user=root
database_password=password
```

```
api_url=http://127.0.0.1/api/v1
callback_url=http://127.0.0.1/callback
```

```
log_server=127.0.0.1
backup_server=127.0.0.1
```

- 3) Use the appropriate command to search for lines containing the word "ERROR" in a log file but exclude lines containing "DEBUG". Save the results to a file named filtered_log.txt.

```
rohit45@rohit45-VirtualBox:~$ vi log.txt
rohit45@rohit45-VirtualBox:~$ cat log.txt
```

```
DEBUG: Starting the application initialization.
ERROR: Unable to connect to the database.
INFO: User 'admin' logged in successfully.
DEBUG: Fetching configuration settings from the server.
ERROR: DEBUG - Configuration settings could not be applied.
INFO: Scheduled job 'backup' completed successfully.
DEBUG: Connection to server timed out. Retrying...
ERROR: Failed to fetch data from API endpoint '/users'.
INFO: Maintenance mode activated.
ERROR: DEBUG - Query execution failed due to a syntax error.
DEBUG: Reloading application modules.
ERROR: Missing required parameter in the request.
INFO: Shutting down the system gracefully.
DEBUG: Closing unused network connections.
ERROR: DEBUG - Unexpected server response received.
```

```
rohit45@rohit45-VirtualBox:~$ cat log.txt | grep -i "error" | grep -vi "debug" >
filtered_log.txt
rohit45@rohit45-VirtualBox:~$ cat filtered_log.txt
```

```
ERROR: Unable to connect to the database.
ERROR: Failed to fetch data from API endpoint '/users'.
ERROR: Missing required parameter in the request.
```

```
rohit45@rohit45-VirtualBox:~$
```

- 4) Write a code to identify the process with the highest memory usage and then terminate it.

```
rohit45@rohit45-VirtualBox:~$ ps aux --sort=-%mem | awk 'NR ==2 {print $2}'
2884
rohit45@rohit45-VirtualBox:~$ kill $(ps aux --sort=-%mem | awk 'NR ==2 {print $2}')
```

- 5) Use the networking tool command and print all the gateway available in a sorted manner

```
vignesh@DESKTOPGV:~$ traceroute -n embedur.ai | awk 'NR>1 {if($2=="*") print $3;else print $2}' | sort
```

```
103.48.181.162
116.119.36.60
160.30.102.74
162.158.52.19
162.159.136.54
172.16.178.1
172.17.0.1
172.23.96.1
182.72.158.41
182.79.161.171
182.79.205.103
192.168.1.1
```

```
rohit45@rohit45-VirtualBox:~$ netstat -rn
```

Kernel IP routing table

Destination	Gateway	Genmask	Flags	MSS Window	irrt lface
0.0.0.0	10.0.2.2	0.0.0.0	UG	0 0	0 enp0s3
10.0.2.0	0.0.0.0	255.255.255.0	U	0 0	0 enp0s3

```
rohit45@rohit45-VirtualBox:~$ netstat -rn | awk 'NR>2 {print $2}' | sort
```

```
0.0.0.0
10.0.2.2
```

```
rohit45@rohit45-VirtualBox:~$
```